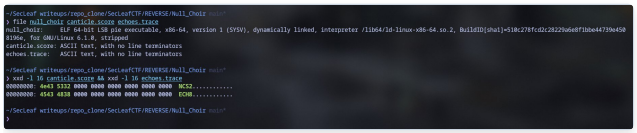
Null Choir

Reverse Engineering



I began by triaging the provided files: stripped ELF64 binary `null_choir`, 196 KB bytecode blob `canticle.score` (magic `NCS2`), and trace blob `echoes.trace` (magic `ECH8`).

Reverse engineering revealed a custom register VM interpreter that maps a 48-byte input across six 64-bit registers, executes 4,096 instructions, and verifies the state against an encrypted target.

I authenticated the 32 genuine records from `echoes.trace` to recover the conductor seed, identified instruction bank 0, and reversed all 4,096 VM instructions and diffusion layers backwards to invert the initial state.

Finally, I tested the recovered flag against the executable / service:

`./null_choir 'SecLeaf{th3_ch01r_s1ng5_1n_5h4d0w_m4ch1n35!!!!!}'`

The program returned Harmony, confirming that the flag was correct.

Flag Captured

`SecLeaf{th3_ch01r_s1ng5_1n_5h4d0w_m4ch1n35!!!!!}`

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